MILITARY SPECIFICATION

HYDRAULIC FLUID, ARRESTING GEAR

This specification has been approved by the Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 This specification covers the requirements for one type and grade of hydraulic fluid containing two corrosion inhibitors, triethanolamine phosphate and sodium mercaptobenzothiazole.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

O-M-232	Methanol (Methyl Alcohol)
PPP-D-729	Drums, Metal, 55-Gallon (For Shipment of Noncorrosive Material)
PPP-P-704	Pails, Shipping, Steel (1 Through 12 Gallons)

STANDARDS

Federal

Federal Test Method Std No. 141	Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling, and Testing

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STANDARDS

Federal (cont'd)

FED-STD-791	Lubricants, Liquid Fuels, and Related Products; Methods of Testing

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

2.2 Other publications - The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials

ASTM D1122-58	Specific Gravity of Concentrated Engine Antifreezes by the Hydrometer
ASTM E70-52T	Determination of pH of Aqueous Solutions with the Glass Electrode

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

3. REQUIREMENTS

3.1 Composition - The composition of the hydraulic fluid furnished under this specification shall be as specified in Table I.

3.2 Components -

3.2.1 Triethanolamine phosphate - The triethanolamine phosphate shall be a mixture of commercially pure triethanolamine and commercially pure phosphoric acid in such proportion that a 10 percent aqueous solution of the mixture shall have a pH value within the range of 7.8 to 8.4. (Any water present in the mixture shall be included in the total percent of water of the hydraulic fluid.)

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TABLE I

Composition

COMPONENT	PERCENT BY WEIGHT	
	MINIMUM	MAXIMUM
Ethylene Glycol	93.95	--
Water	2.5	3.0
Triethanolamine Phosphate	2.25	2.75
Sodium Mercaptobenzothiazole	0.2	0.3

3.2.2 Sodium mercaptobenzothiazole - The sodium mercaptobenzothiazole content of the finished hydraulic fluid shall conform to the requirements of Table I when tested as specified in 4.3.2. Sodium mercaptobenzothiazole, when blended with the other components, shall be in the form of an aqueous solution consisting essentially of sodium mercaptobenzothiazole and water. The aqueous solution shall contain not less than 45 percent by weight of sodium mercaptobenzothiazole when tested as specified in 4.3.1.1. Chemically pure sodium sulfite, when used as a stabilizer, shall not exceed one percent by weight of the formulated sodium mercaptobenzothiazole solution. The aqueous solution shall be known as "NaMBT inhibitor," and shall conform to the requirements of Table II.

TABLE II

Chemical and Physical Requirements of NaMBT Inhibitor

REQUIREMENT	LIMITS	TEST PARAGRAPH
Color	Not darker than 15	4.3.1.2
Specific Gravity at 25°/25°C (77°/77°C)	1.27 ± 0.03	4.3.1.3
Suspended matter	None	4.3.1.4
Solubility at 25 ± 5°C (77 ± 8°F) in:		
Water	Soluble; max slight haze	4.3.1.5
Ethylene Glycol after 24 hrs	Soluble; no haze or cloudiness	4.3.1.6
Water and Ethylene Glycol	Soluble; max slight opalescence	4.3.1.7
Free Alkalinity (Calculated as grams Na OH per 100 mls at 25 ± 3°C (77 ± 5°F)	0.05 to 0.75	4.3.1.8

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3.3 Finished hydraulic fluid - The finished hydraulic fluid shall conform to the requirements specified in Table III.

TABLE III

Chemical and Physical Requirements of Finished Hydraulic Fluid

REQUIREMENT	LIMITS	TEST PARAGRAPH
Phosphate Content, percent minimum (Calculated as phosphoric acid (H_3PO_4))	0.56	4.3.3
pH value, 50 percent aqueous solution (by volume) at 25 ± 3°C (77 ± 5°F)	7.2 to 7.8	4.3.4
Boiling point, minimum	165°C (329°F)	4.3.5
Ash, percent by weight, maximum	0.52	4.3.6
Specific Gravity of undiluted material at 60°/60°F	1.111 to 1.123	4.3.7
Viscosity in Centistokes at 100°F (37.7°C) minimum	9.0	4.3.8

3.4 Workmanship - The hydraulic fluid shall be manufactured by such processes as to produce a homogeneous and uniform product free from suspended material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling -

4.2.1 Lot - All material manufactured as one batch and offered for delivery at one time shall be considered a lot for purposes of acceptance inspection and tests.

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4.2.2 Sampling for tests - A sample of finished hydraulic fluid shall be selected at random in accordance with Method 8001 of Standard Federal Test Method Standard No. 791 (ASTM-D270) and subjected to all the applicable tests of this specification as described under "Test Methods." If the sample fails any of the tests, the lot represented by the sample shall be rejected.

4.2.2.1 NaMBT inhibitor - An 8-ounce sample of the NaMBT inhibitor used in the formulation of each lot shall be furnished with the sample of the finished hydraulic fluid. The sample shall be subjected to all the applicable tests as described under "Test Methods." If the sample fails any of the tests, the lot represented by the sample shall be rejected.

4.2.3 Sampling for inspection of filled containers - A random sample of filled containers shall be selected in accordance with Standard MIL-STD-105. The inspection level shall be I and the acceptable quality level (AQL) shall be 2.5 percent defective. Each filled container selected shall be examined to verify compliance with all requirements of this specification in regards to fill, closure, marking, and other requirements not involving tests.

4.3 Test methods -

4.3.1 NaMBT inhibitor -

4.3.1.1 Sodium mercaptobenzothiazole content of NaMBT inhibitor - Weigh (to the nearest 5 milligrams) a 5.000 to 5.200 gram sample of NaMBT inhibitor in a clean, tared 250-ml glass beaker. Add approximately 100 milliliters of distilled water. Add CP concentrated hydrochloric acid cautiously, with constant stirring, until the solution is just acid to litmus, and then add 10 drops of the 12N hydrochloric acid in excess. Stir the mixture vigorously with a glass stirring rod for approximately one minute. Cool the mixture to a temperature of 10° to 15°C (50 to 59°F) and allow it to stand at this temperature for one hour. Prepare a filter consisting of a Gooch crucible and a mat of asbestos, and dry it in an oven at 105 to 110°C (221 to 230°F), to constant weight. Filter the mixture through the tared Gooch crucible, using suction. Wash the precipitate in the crucible with five 10-ml portions of cold distilled water, allowing each portion of wash water to pass through the filter before adding the next portion. Remove as much of the last portion of wash water as possible by suction. Dry the precipitate (in the crucible) for three hours in an oven at 105° to 110°C (221° to 230°F). Cool the crucible and contents for at least 15 minutes in a desiccator, and weigh. Repeat the drying and weighing to constant weight. Calculate the percent sodium mercaptobenzothiazole (NaMBT) as follows:

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$$\text{Percent NaMBT by weight} = \frac{P\ (1.1316)}{W} \times 100$$

Where: P = Weight in grams of precipitate
W = Weight in grams of sample

4.3.1.2 Color - A representative sample of NaMBT inhibitor shall be examined visually in a Hellige Color Comparator, or equivalent. The result shall be expressed as a number in the scale of the Gardner Color Standards of 1953. An approximate conversion of color designation from one system to another can be accomplished by means of the Color comparison chart in accordance with Method 4249 of Federal Test Method Standard No. 141.

4.3.1.3 Specific gravity - The specific gravity of the NaMBT inhibitor shall be determined in accordance with ASTM Method D1122-58.

4.3.1.4 Suspended matter - Suspended matter in the NaMBT inhibitor shall be determined by inverting a bottle of the sample and examining by transmitted light.

4.3.1.5 Solubility of NaMBT inhibitor in water - Dilute a portion of the inhibitor with distilled water to form a solution containing 0.2 percent by weight of sodium mercaptobenzothiazole. Not more than a slight haze shall be permitted after standing 24 hours at 25°C (77°F).

4.3.1.6 Solubility of NaMBT inhibitor in ethylene glycol - The inhibitor shall be completely soluble in ethylene glycol containing 2.5 percent by weight of triethanolamine phosphate, when added in the ratio of three parts by weight of NaMBT inhibitor to 100 parts by weight of the ethylene glycol-triethanolamine phosphate mixture and thoroughly mixed. The mixture shall not develop any haze or cloudiness after being allowed to stand undisturbed and protected from direct sunlight at a temperature of 25 ± 5°C (77 ± 8°F) for 24 hours.

4.3.1.7 Solubility of NaMBT inhibitor in water and ethylene glycol mixture - Prepare a solution by thoroughly mixing 0.134 gram of NaMBT inhibitor with 30 milliliters of ethylene glycol containing 2.5 percent by weight of triethanolamine phosphate and having a pH value of 7.2 to 7.8. Dilute with distilled water to 100 milliliters and allow to stand 16 hours. The solution shall show no more than a barely perceptible opalescence and not more than a trace of precipitate.

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4.3.1.8 Free alkalinity of the NaMBT inhibitor - Transfer 10 mls of the sample to a clean 250-ml glass beaker by means of a 10-ml pipette. Precautions should be taken to obtain only the clear solution for the analysis. Prepare a mixture consisting of equal parts by volume of uninhibited ethylene glycol and distilled water. Add 100 mls of the glycol-water mixture to the sample of "NaMBT" in the beaker. Titrate the solution potentiometrically, using a pH meter equipped with a glass electrode and a suitable reference electrode, to a pH value of 9.0 ± 0.05, using standardized N/10 or N/20 hydrochloric acid. During the titration, the solution shall be stirred effectively. A glass stirrer, either electric or air driven may be used. Stirring shall not be accomplished by blowing air through the solution as this tends to oxidize the "NaMBT." The free alkalinity shall be calculated as grams of NaOH per 100 ml of sample.

$$\text{grams of NaOH per 100 ml of sample} = \frac{AB\ (0.040)}{C} \times 100$$

Where: A = ml of HCl
B = normality of HCl
C = ml of sample

4.3.2 Sodium mercaptobenzothiazole content of finished product - Transfer a 10.00 to 12.00 gram sample of the NaMBT inhibited hydraulic fluid into a clean tared 250-ml Erlenmeyer flask. Turbid samples should be filtered before weighing. Add 10 mls of Grade A synthetic methanol conforming to Specification O-M-232 and mix the liquids by shaking the flask. A 20-gram sample and 15-ml of methanol should be used when the hydraulic fluid contains less than 50 percent of ethylene glycol. Methanol is used to keep the NaMBT in solution. Add dilute (approximately 3N) hydrochloric acid dropwise, shaking the flask frequently to mix the contents, until the pH value of the solution is 4.0 ± 0.2, as indicated by appropriate indicator paper or by a pH meter. Add exactly 20 mls of standardized N/50 iodine solution, and mix well by shaking the flask. Adjust the temperature of the mixture to 25 ± 5°C (77 ± 8°F), and allow the flask to stand at this temperature for approximately 15 minutes to insure complete reaction of the iodine with NaMBT. Next, add approximately 100 mls of distilled water; mix well by shaking the flask and adjust the temperature of the mixture to 25 ± 5°C (77 ± 8°F). Titrate the excess (unconsumed) iodine with standardized N/50 sodium thiosulfate solution, using starch as an indicator. Calculate the percent NaMBT by weight as follows:

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$$\text{Percent NaMBT by weight} = \frac{(A)\ (0.1892)\ (100)}{W}$$

Where: A = ml of N/1 iodine consumed by sample = (ml of I_2 x normality) - (ml of $Na_2S_2O_3$ x normality)
W = weight of sample (grams)

4.3.3 Phosphate content -

4.3.3.1 Preparation of magnesia mixture - Dissolve 50 g of crystalline magnesium chloride ($MgCl_26H_20$) and 100 g of ammonium chloride in 500 ml of water. To this solution add about 10 ml of concentrated aqueous ammonia, allow to stand overnight, and filter if a precipitate appears. Make this solution slightly acid with hydrochloric acid, dilute to one liter, and keep in a glass stoppered bottle.

4.3.3.2 Procedure - Dilute a 20-gram sample, weighed to the nearest 0.01 gram, of the finished product with 100 ml of distilled water. To this mixture add 10 ml of a mixture of equal parts by volume of concentrated hydrochloric acid and water, and 30 ml of magnesia mixture. Add concentrated aqueous ammonia until the solution is neutral to methyl red indicator. Stir for five minutes, and while continuing to stir add 15 ml additional aqueous ammonia. Allow the solution to stand for four hours or longer, filter, and ignite the precipitate in a weighed platinum crucible at 1000 to 1050°C (1832 to 1922°F) to constant weight. Calculate the phosphate concentration from the weight of the magnesium pyro-phosphate ($Mg_2P_2O_7$).

4.3.4 pH value - A 50 percent aqueous solution shall be prepared by mixing equal volumes of the finished hydraulic fluid and distilled water. The pH value shall be determined in accordance with ASTM Method E70-52T.

4.3.5 Boiling point - One hundred ml of the hydraulic fluid, together with a few chips of unglazed porcelain or other suitable material to prevent superheating, shall be placed in a 200 ml round bottom flask fitted with a reflux condenser. The innertube of the condenser must be wide enough to accommodate a thermometer safely. A thermometer shall be passed through the reflux condenser by means of a long, thin wire in such a way that the bulb of the thermometer shall be completely immersed in the liquid. No part of the bulb of the thermometer shall be closer than 1/2 inch to the bottom of the flask, and no part of the thermometer shall be in contact with the flask. The liquid shall be heated on a hot plate with refluxing until the equilibrium

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temperature is reached. At this point the contents of the flask shall be swirled around in such a way that the moisture condensed on the walls of the flask shall be washed down with hydraulic fluid. The temperature at which the liquid boils immediately after this treatment shall be taken as the boiling point of the product.

4.3.6 Ash - The ash content of the final product shall be determined by evaporation of the major portion of the glycol and ignition of the residue in accordance with the following procedure: Ignite a clean size 1A glazed porcelain or vycor crucible containing approximately 1.0 gram of analytical grade shredded asbestos in a muffle furnace for 1 hour at a temperature of 875 ± 15°C (1607 ± 27°F). Cool the crucible and contents for 10 to 15 minutes in the open air, then for 15 to 20 minutes in a desiccator, and weigh on an analytical balance. Add from 5 to 7 grams of the hydraulic fluid to the crucible; and immediately weigh the crucible and contents as rapidly as possible on an analytical balance. Evaporate the hydraulic fluid for a period of four hours at a temperature of 160 to 190°C (320 to 374°F) on an electric hot plate or by means of an "Infraradiator" or comparable apparatus. Control the temperature so that there is no loss of material by spattering during the evaporation. Next, place the crucible in a muffle furnace at a temperature not higher than 200°C (392°F); and gradually raise the temperature to 875 ± 15°C (1607 ± 27°F) over a period of two hours. Maintain the temperature at 875 ± 15°C (1607 ± 27°F) for a period of one hour. Cool the crucible and contained ash for 10 to 15 minutes in the open air, then for 15 to 20 minutes in a desiccator; and weigh rapidly on an analytical balance. Calculate the percent of ash as follows:

$$\frac{\text{Weight of ash}}{\text{Weight of sample}} \times 100 = \text{percent ash}$$

The asbestos is used to prevent loss of sample by spattering during evaporation and ignition. Satisfactory results cannot be obtained by direct ignition of glycol over an open flame. Rapid weighing is required because of the hygroscopic nature of glycol and of the ash.

4.3.7 Specific gravity - The specific gravity of the finished hydraulic fluid shall be determined in accordance with ASTM Method D-1122-58.

4.3.8 Viscosity - The kinematic viscosity of the finished hydraulic fluid shall be determined in accordance with Method 305.4 of FED-STD-791.

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4.4 Rejection criteria - When any acceptance test sample fails to meet any of the test requirements of this specification or when the number of defective filled containers exceeds the acceptance number as specified in 4.2.3, the lot represented by the sample shall be rejected.

5. PREPARATION FOR DELIVERY

5.1 Packaging and preservation - Not applicable.

5.2 Packing - Packing shall be level A or C as specified (see 6.2).

5.2.1 Level A - The hydraulic fluid shall be furnished in 5-gallon or 55-gallon containers as specified (see 6.2). The 5-gallon containers shall conform to Type I (screw-cap closure), Class 3 or 8 of Specification PPP-P-704 and shall be provided with inner seals and push-pull spouts. The 55-gallon drums shall be closed-top drums, Type I or II, conforming to Specification PPP-D-729.

5.2.2 Level C - The hydraulic fluid shall be packaged to afford the minimum degree of protection necessary to prevent deterioration or damage during shipment under normal environmental conditions and commercial modes of transportation.

5.3 Marking - In addition to any special marking required by the contract or order, each container shall be marked in accordance with the requirements of Standard MIL-STD-129 and shall include the warning markings as specified in 5.3.1.

5.3.1 Warning marking - The following warning marking shall appear on each container:

> WARNING: This product while undiluted will freeze at about 0°F. For use at lower temperatures water must be added as required to obtain lower freezing point.

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6. NOTES

6.1 Intended use - The hydraulic fluid covered by this specification is intended for use in aircraft arresting gear systems. This fluid is not interchangeable with any other type or grade of hydraulic fluid.

6.2 Ordering data -

(a) Title, number, and date of this specification
(b) Type and size of containers (see 5.2)
(c) Quantity
(d) Level of packing required (see 5.2)

6.3 Basis of purchase - The hydraulic fluid covered by this specification should be purchased by volume, the unit being one U. S. gallon, 231 cubic inches at 15.5°C (60°F). The weight per gallon shall have an approximate range from 9.25 to 9.35 pounds.

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a. Paragraph Number and Wording:

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